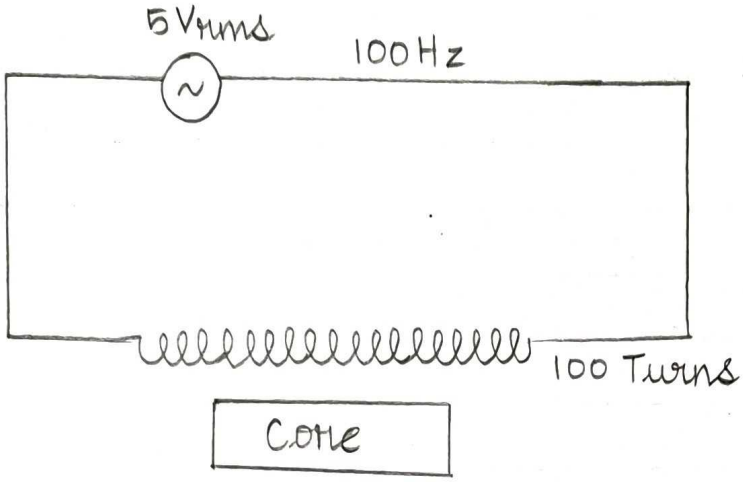
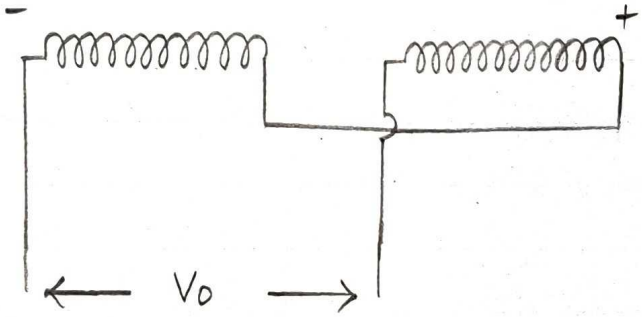
CIRCUIT DIAGRAM





TABULATION

	Input (v) voltage (V_m)	Output voltage	frequency (Hz)	Time = $1/f$
Positive displacement	7	2.2 V (0 = 5mm)	1000	$\frac{1}{1000}$ = 1ms
Negative displacement	7	(0 = -5mm)	1000	1ms

CHARACTERISTICS OF LVDT
(LINEAR VARIABLE DIFFERENTIAL TRANSFORMER)

AIM:

To understand and simulate the relation between and displacement and output voltage.

APPARATUS REQUIRED:

Laptop with internet connection.

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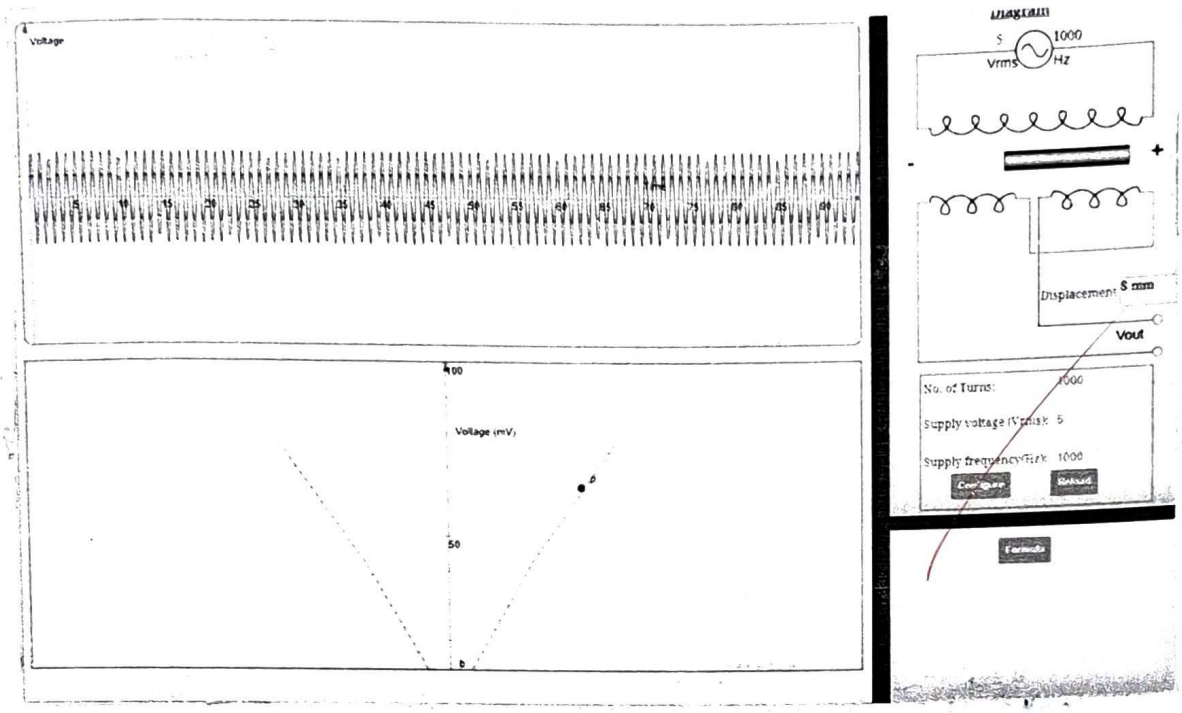
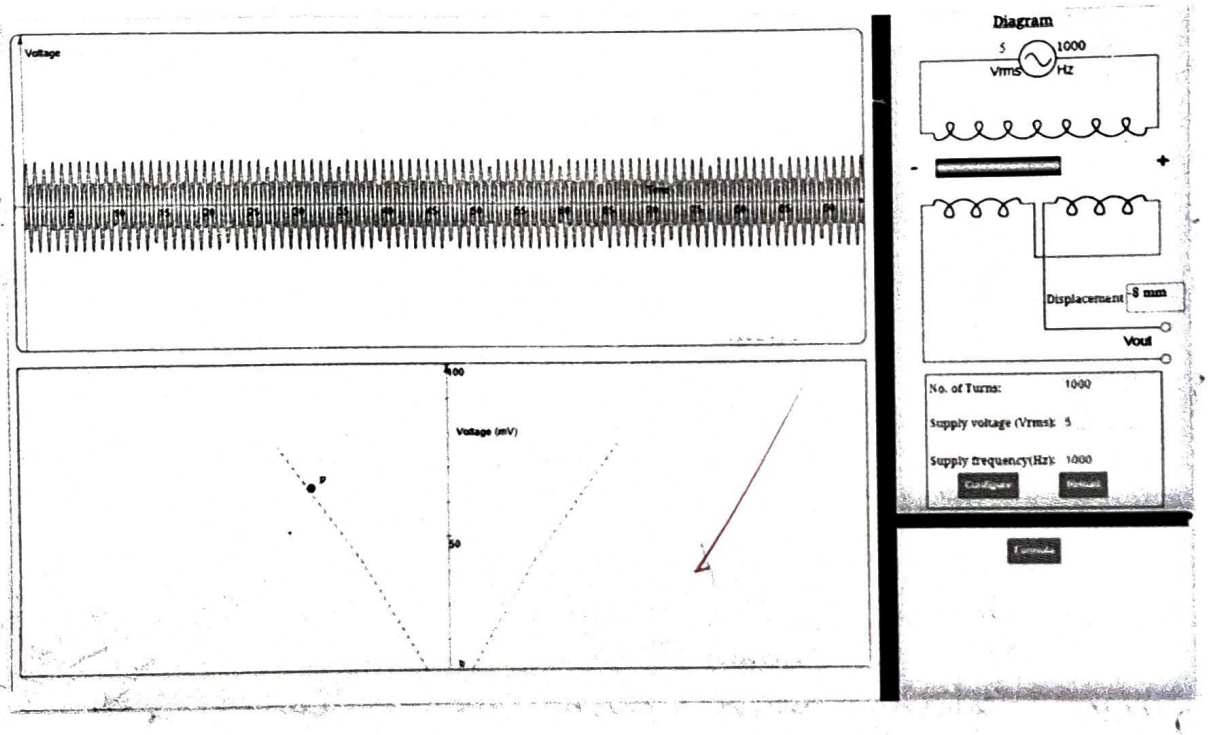
THEORY:

LVDT is an inductive transducer that converts linear displacement into a electrical signal. It consists of a transformer having one primary winding and two secondary windings wound on a core. As the primary is connected to an AC source the AC current and the voltages are produced in the secondary of LVDT

When the core is at the center, the flux linking with both the secondary winding are equal this means there is no displacement when the core is at the right,


Voltage
Time
400
Voltage (mV)
50
Diagram
5
Vrms
1000
Hz
-
+
Displacement 5 mm
Vout
No. of Turns: 1000
Supply voltage (Vrms): 5
Supply frequency(Hz): 1000
Configure
Reset
Formula


Voltage
Time
400
Voltage (mV)
50
Diagram
5
Vrms
1000
Hz
-
+
Displacement -8 mm
Vout
No. of Turns: 1000
Supply voltage (Vrms): 5
Supply frequency(Hz): 1000
Configure
Reset
Formula

the flux linking with the secondary winding S_1 is more. So, the emf induced in S_1 is more than S_2. So, the net emf is positive. When the core is at the left, the flux liking and the emf in S_2 is more than S_1. So, the net emf is negative.

PROCEDURE:

* Connections are made as per the circuit diagram.

* Set the number of turns, supplied voltage and supplied frequency

* Configure the parameters

* Move the core to positive side and the negative side and observe the input voltage and output voltage waveforms.

* Plot the graph between input and output voltage.

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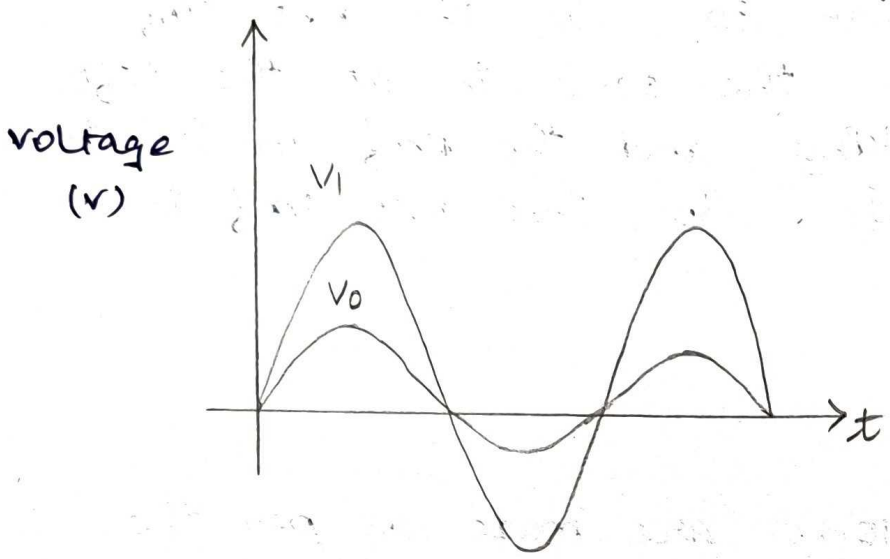
MODEL GRAPH

Positive displacement



calculation

$V_m = \sqrt{2} \times V_{rms}$

$= \sqrt{2} \times 5$

$= 7V$

$T = 1ms$

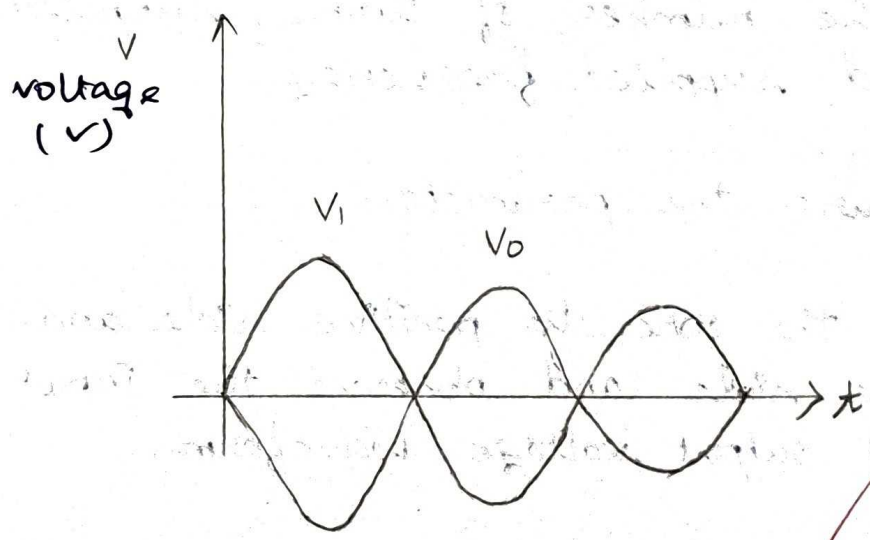
$4SD = T$

$1SD = T/4$ V

$1.25SD = T/4 \times 1.25$

$= 2.2V$

Negative displacement



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RESULT:

To understand and simulate the relation between core displacement and output voltage were developed and tested successfully.